Mean test, σ unknown -> Student t + normality checks

A1S2: SLIDES 1-51 ONLY

1 RECOGNISE IT

- Hypothesis about mean μ; population σ unknown.
- Raw sample or $\bar{X}$,s,n supplied.
- Need to justify the t lane and inspect normality.

2 DECISION / METHOD

1. Compute n, $\bar{X}$, s.
2. Check random sampling and approximate population normality.
3. Use tcalc with SE=s/√n.
4. df=n-1.
5. Continue with critical or p-value route.

3 FORMULA

$t_{calc}=(\bar{X}-\mu_0)/(s/\sqrt{n})$
$df=n-1$

4 FORMULA SHEET VS WHAT I KNOW

Gives: $t_{(n-1)}$ and $s/\sqrt{n}$ when σ unknown.
Still know: normality evidence: symmetry, histogram/boxplot and linear normal probability plot.

5 EXCEL ROUTE

- n=COUNT(range)
- Xbar=AVERAGE(range)
- s=STDEV.S(range)
- tcalc=(Xbar-mu0)/(s/SQRT(n)); df=n-1.

6 TINY WORKED EXAMPLE

$\bar{X}=31.5, s=6, n=16, \mu_0=30$ -> SE=1.5, tcalc=1.0, df=15. Normality should be checked because the sample is small.

7 TOP EXAM TRAPS

- Using Z with sample s.
- df=n.
- Treating a single outlier-free boxplot as proof.
- Forgetting the population-normality condition for small n.

8 SAY ALOUD

Unknown σ sends the mean test to Student t with sample s, n-1 df, and a normality check.

9 EXIT CHECK

What three Excel summaries are needed before a raw-data t test?

ONE-GLANCE VISUAL ROUTE

μ test > σ unknown > normality check > s/√n, df=n-1 > Student t